

TIPPING FRONTIER CONSULTING REPORT

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Table of Contents

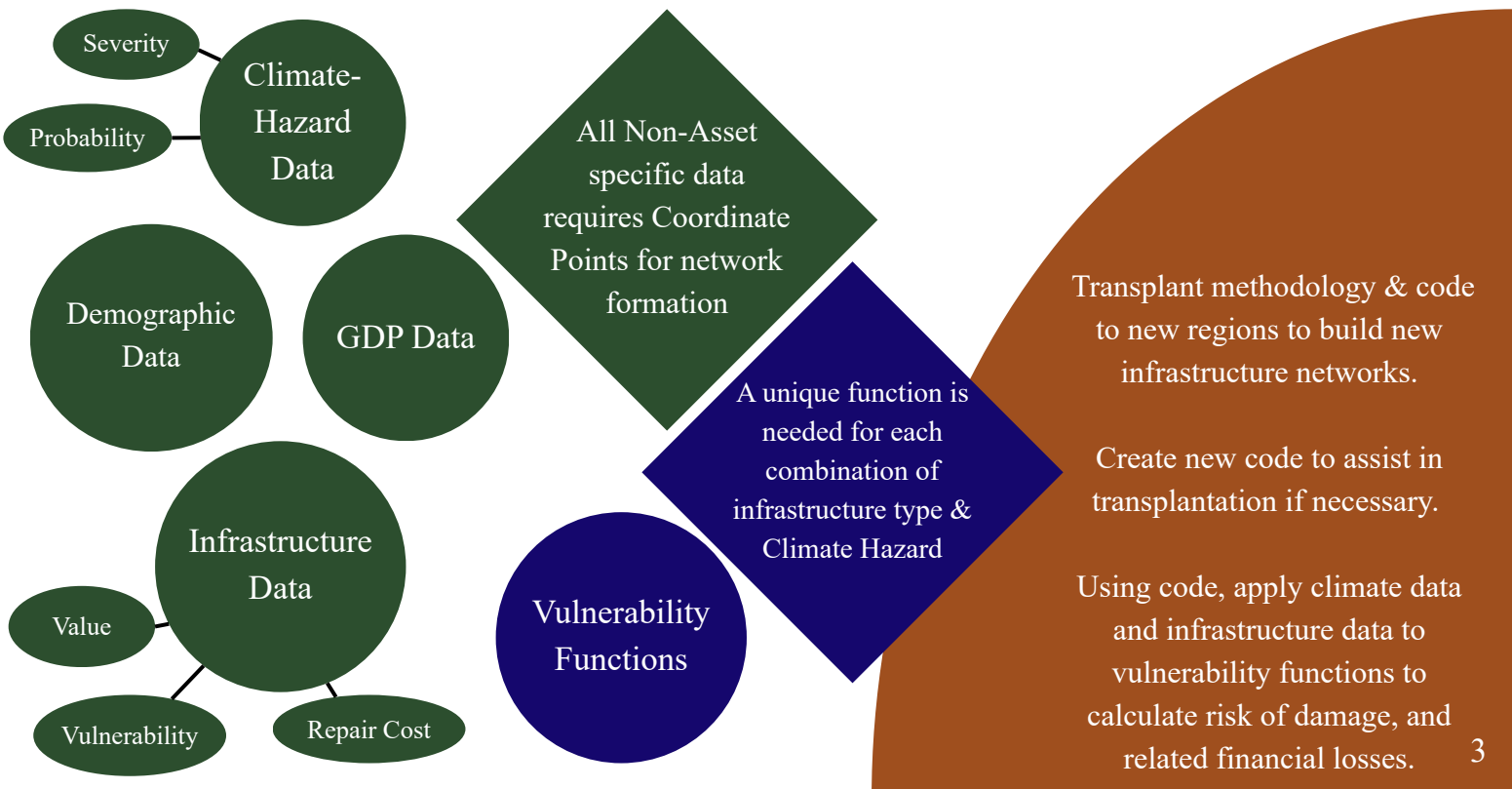
<u>1. What is Tipping Frontier?</u>	p.3
<u>2. Project Scope and Objectives</u>	p.4
2.1 Our Tasks and Project Structure	
2.2 Relevance for Tipping Frontier and Clients	
<u>3. Task 1 – India’s Electricity Grid</u>	p.5
3.1 Scope and Goal	
3.2 Data	
3.3 Methodology	
3.4 Challenges and Responses	
3.5 Model Limitations	
3.6 Conclusion	
<u>4. Task 2 – Ugandan Hydroelectric Powerplants</u>	p.15
4.1 Scope and Goal	
4.2 Conceptual Framework	
4.3 Assets Considered	
4.4 Hazards Considered	
4.5 Methodology	
4.6 Results and Implications	
4.7 Challenges	
4.8 Limitations and Relevance	
4.9 Conclusion	
<u>5. Final Summary and Project Relevance</u>	p.24
<u>Bibliography</u>	p.25
<u>Appendix</u>	p.26-29

Tipping Frontier (TF) is a financial consulting company modelling climate, pandemic and conflict risk for institutional investors and public-sector clients. TF uses open-source data and existing methodology to make in-house models demonstrating how these risks may affect infrastructure and economic activity in a given economy. These models are then employed as decision-support tools used to advise clients on investment strategies, intervention and mitigation opportunities, and the development of financial products.

For financial institutions, TF’s analysis is primarily used to inform investment and asset-management decisions by identifying climate-exposed assets, quantifying downside risk, and comparing relative risk–return profiles. While historically framed around risk avoidance, climate-resilient investment is increasingly viewed as both a risk-management tool and a source of long-term value. Governmental clients typically have broader mandates, combining the protection of livelihoods and the reduction of systemic risk with the need to ensure economic stability and continuity. Reflecting these different objectives, TF’s models are applied flexibly: some clients use them to reallocate capital away from high-risk assets, while others use them to assess existing exposure and evaluate adaptation/mitigation options. In practice, this can range from asset-level risk screening to advising on collective resilience measures, such as coordinating investment in shared flood protection infrastructure, to reduce vulnerability across multiple stakeholders.

Tipping Frontier works with both domestic and international clients. Its long-term objective is to build a comprehensive climate-risk “universe”: a unified modelling framework covering multiple countries, key infrastructure systems (e.g. power, transport), and a wide range of climate hazards. The aim is to centralise climate-risk intelligence, reduce duplication of effort across projects, and provide clients with a consistent and scalable decision-support tool.

This is an inherently ambitious undertaking. Even modelling a single country completely requires the integration of the following:



TF had already developed substantial internal resources, including infrastructure vulnerability data and repair cost assumptions across asset classes. In particular, their internal library contained a detailed methodological document outlining how climate hazards are translated into physical damage and GDP-related losses.

However, the application of this methodology to a new country had not yet been completed and required significant upfront work: compiling and validating region-specific datasets, adapting network representations, and ensuring that vulnerability assumptions remain credible in different physical and institutional contexts. As TF is still a relatively new organisation, much of this work was ongoing. The underlying frameworks, data, and plans existed, but had not yet been fully operationalised into reproducible, end-to-end outputs.

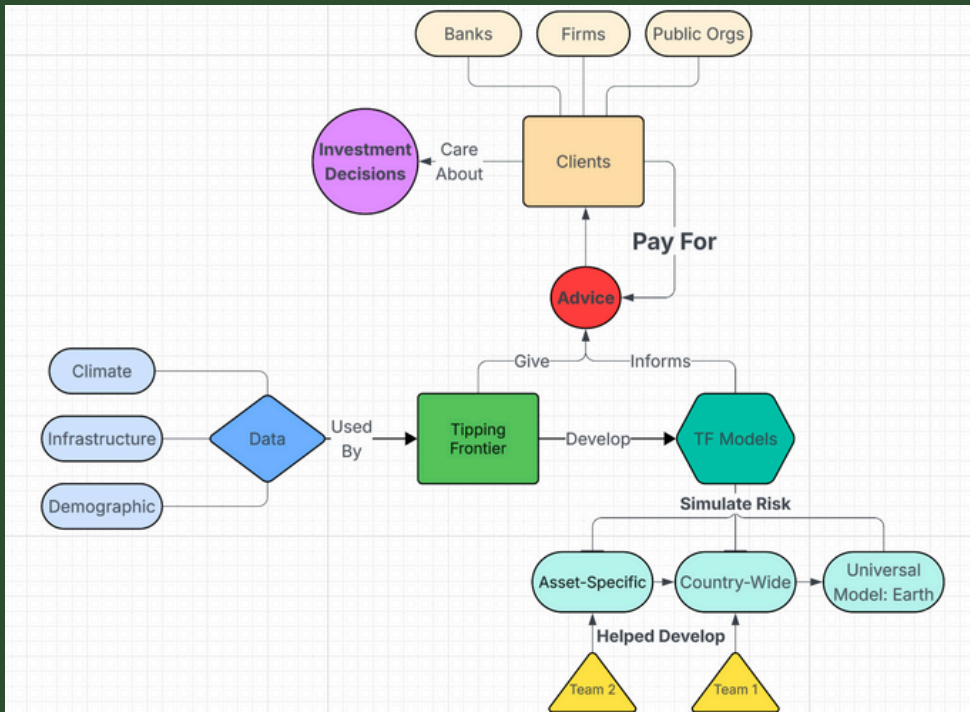
In the initial project brief, TF indicated that our work could involve either model prototyping with full documentation of assumptions and code, or structured research and reporting. Beyond this, the scope was intentionally open-ended.

One of the first steps of the project therefore consisted of clarifying objectives jointly with the client. Early discussions revealed that TF was preparing for prospective client engagements in India and Uganda, but had not yet completed their goals for these countries.

As a result, the project developed around supporting TF’s ongoing business development by operationalising their modelling capabilities in these two imminently relevant contexts. Our team was divided into two groups, each working on a different but complementary component of TF’s modelling pipeline. Team 1 focused on completing the electricity network layer of the climate-hazard risk model for India, adapting an existing South-East Asia framework to a new system. Team 2 conducted an asset-level climate risk assessment of three hydropower plants in Uganda, providing a prototype for more granular, asset-specific analysis that goes beyond national or sector-level screening.

Both work-streams were explicitly practical and model-oriented. The objective was not to provide investment recommendations, but to expand TF’s internal modelling capacity, validate key components of its methodology in new settings, and identify where assumptions, data constraints, or computational limits become binding. In doing so, the work helps TF reduce lead time for future client engagements, demonstrate methodological robustness in new geographies, and lay some modelling groundwork for TF to build onto for credible risk insights to clients operating in climate-exposed infrastructure markets. Both teams worked on new and unfinished projects. Therefore, our work was crucial by serving as proof-of-concepts; we turned ideas and frameworks into real output.

Figure 1:
Flowchart
illustrating TF
Business Model and
Team relevance



Scope and Goal

TF initially tasked us with translating their South East Asia climate-risk model to India in order to expand the model's "universe". The existing SEA prototype consisted of multiple layers, including an infrastructure layer (rail, roads, and power lines), a hazard layer (such as heavy winds, monsoon activity, and earthquake probability), and exposure layers calculating GDP at risk based on infrastructure characteristics (value, vulnerability, time to repair) and hazard characteristics (severity and probability). In practice, this scope proved too ambitious for the duration of our contract, particularly given our initial unfamiliarity with the full methodology. As a result, an early priority was to clarify what expanding the model's "universe" would realistically involve.

Through a series of meetings with TF, we focused on identifying where support was most needed and which components of the model were not yet functional. This process highlighted four key problem areas: (1) inadequate data visualisation, (2) missing vulnerability functions for new hazards, (3) a non-functional network construction pipeline, and (4) incomplete calculation of GDP losses.

Following further discussion, it was agreed that our work would focus on rebuilding the electricity transmission network, as this was both a core infrastructure layer and a clear bottleneck in the India climate-hazard risk model. We used these discussions to narrow the scope of work and define a plan that would deliver a concrete, achievable output aligned with TF's priorities. The resulting deliverable was a self-contained codebase that maps India's power grid and calculates GDP exposure along transmission lines, producing a prototype suitable for demonstration to prospective clients. This component forms part of the broader infrastructure layer of TF's India climate-risk model and directly addresses two of the identified problem areas: network construction and GDP exposure calculation.

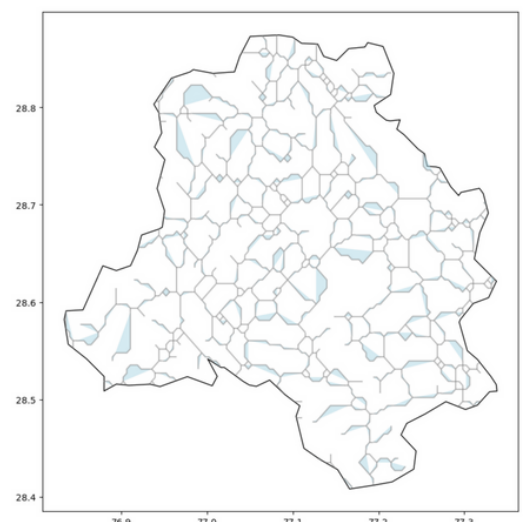
Although previous interns had produced code to generate the power network and visualise GDP exposure, it was not documented and difficult to extend. Team 1 therefore decided to rebuild the component from scratch, allowing us to develop a clear understanding of the model and produce an implementation that could be explained, maintained, and built upon going forward. TF were initially somewhat unclear about the nuances of this task, and so our meetings were mostly structured in a semi-informal sense. We raised questions and discussed all points of clarification, showing any working code with them. Over time it became much more clear what the end result would look like.

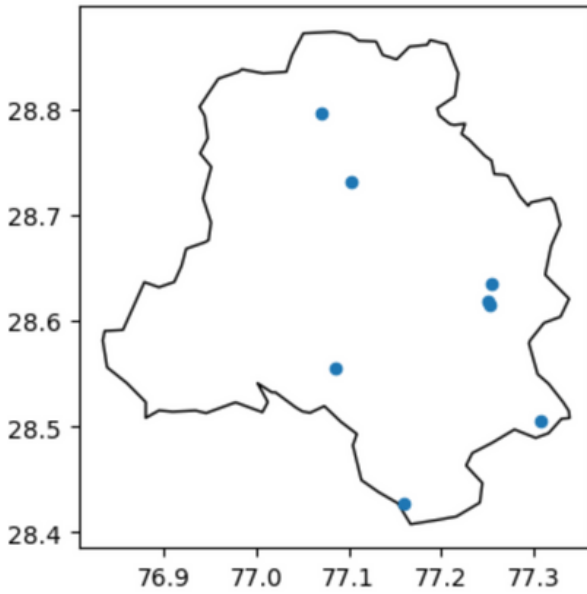
Data

This section is intended to give an overview of the different datasets we were given to use, the challenges each of them pose and how and why they are involved in the final model.

GRID

The grid data (Ardene et al., 2020) provides a representation of the electric transmission lines covering our geographic area, a visual aid of Delhi's grid data is as shown. This data is essential for the subsequent analysis aimed at identifying transmission lines of varying strategic importance. Grid data is represented in a geopackage file, using what is called "linestring" data, geometric data that stores coordinates for lines, rather than points or shapes.





POWERPLANTS

Power plant data (Resource Watch, n.d) represents where each of the power plants in our geographic area is, as well as crucial information such as their fuel capacity and power generation for different years.

This, along with target data, can then be used to predict how much electricity each electrical transmission line holds. Again, a visual aid is provided of the power plants corresponding to the Delhi region.

Power plant data are stored in a CSV file containing numerous attributes that characterise each facility, including geometric point data specifying geographic coordinates. Due to the centralised nature of electricity generation and the necessity for distribution across the network, the spatial density of power plant locations is lower than that of target nodes or transmission line infrastructure.

TARGETS

Within the model, targets (Ardene et al., 2020) represent grid cells of approximately $742 \text{ m} \times 742 \text{ m}$ covering the geographic area of India. Each cell represents a location that is connected to the national electricity grid. Non-targets are usually natural areas, lakes, or places not requiring electricity.

The data is represented as a binary raster, with a series of geometric points corresponding with a binary value. '1' signalling a connected area to the electricity grid, and '0' a non-connected area.

GDP

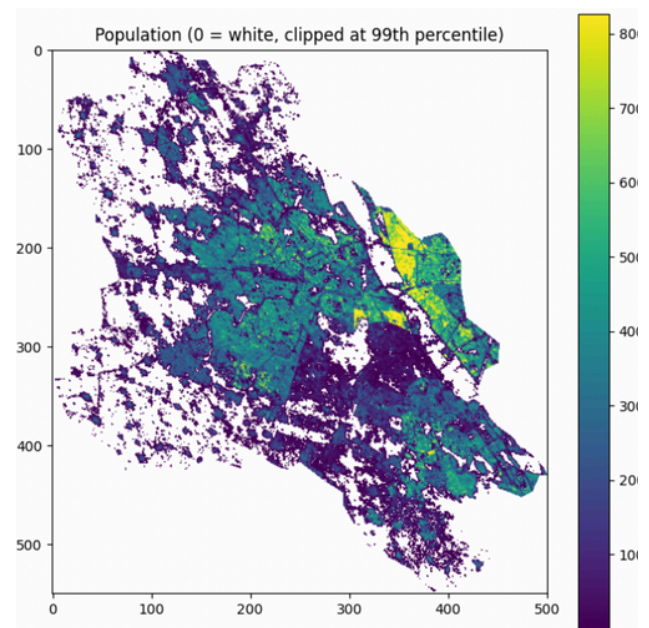
GDP is Tipping Frontier's primary method for assigning value to different parts of the network. The metric acts as a proxy reflecting economic activity levels. This is appropriate as TF's clients are often financial institutions. In our model we primarily assign GDP values for targets and grid lines. We were given two sources to obtain this data. One was a geopackage file containing different values representing GDP, corresponding to geometric "shape" data, used to represent different regions of about 9.28km squared radius. The other was a geopackage file containing values representing GDP per Indian district. With the latter file, an attempt was made to divide GDP by population, to create a unique GDP per capita dataset. However, it was found that it was better to use the former file, as trying to create a new dataset added little in terms of granularity while considerably increasing complexity and computational load.

(European Commission Joint Research Centre, 2023)

POPULATION

Population is used as a secondary method to assign value to assets, and is also used to calculate exact GDP values for targets and grid lines, because, as previously mentioned, our GDP data is represented in a GDP per capita file.

This data is represented in a raster file with coordinate points in the Molleweide coordinate system corresponding to the number of people who reside in a given 100m^2 area. A visualisation for Delhi is provided to the right. Note that these population values are float values rather than integers. The base population file uses float to allow for population values (eg. 65) to be split between smaller geographical areas (32.5 per area), even though it implies a 'part' of a person. (Byers et al., 2019)



CAPACITY

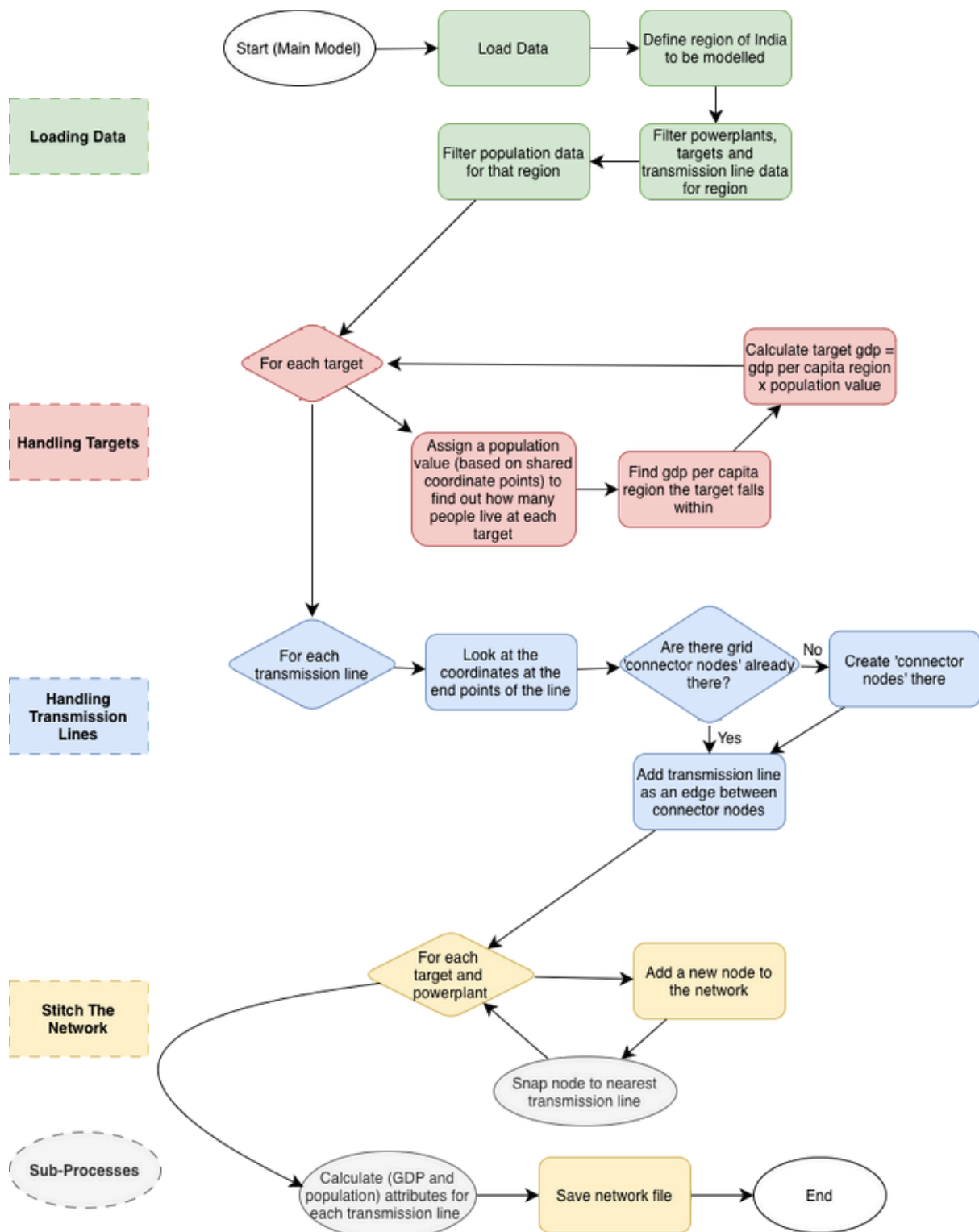
Capacity is an important attribute used to assign value to assets. As previously mentioned this data is contained within the power plant file already, so no additional files were required.

Methodology

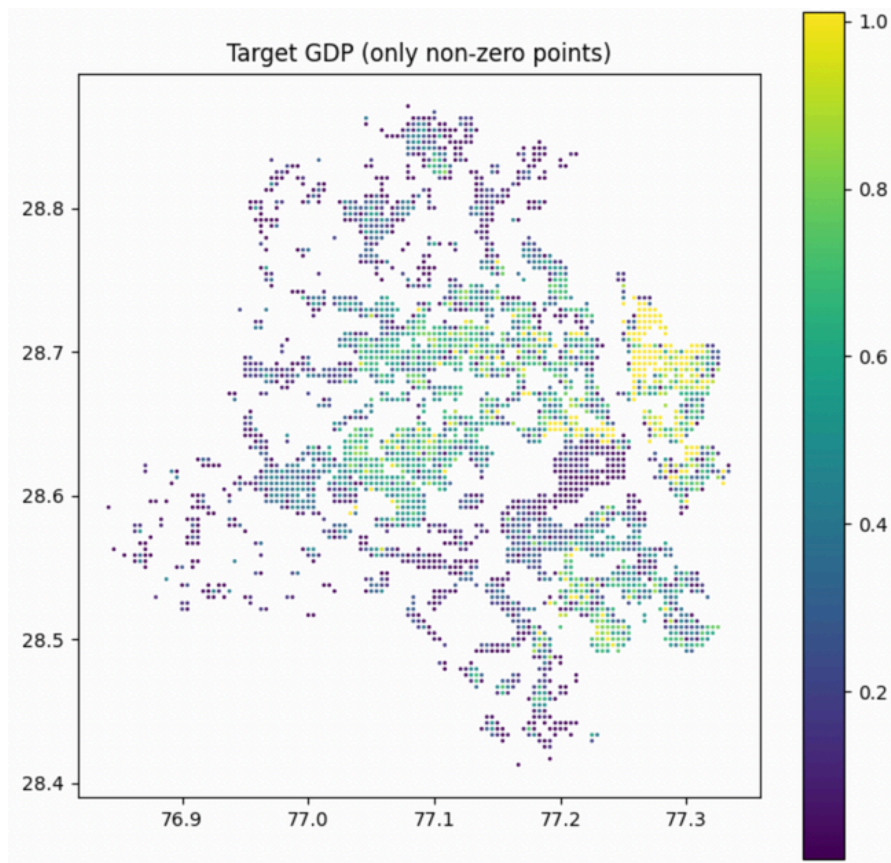
As alluded to previously, we were given 2 prior projects as reference material to assist us with coding this project. The first is 'open GIRA' (General Model for Incident Risk Analysis) (Couce-Vieira et al., 2017), an open-source GitHub repository created by NISMOD (National Infrastructure Systems Model) an organisation operating within OPSIS (Oxford Program for Sustainable Infrastructure Systems). The second reference project is 'seasia', a direct application of climate risk modelling on South East Asia, created by Oxford Infrastructure Analytics.

After careful study of the reference material along with discussion and clarification with our client, our final model is communicated through the flowcharts as seen below. In the following section we detail the challenges we encountered in building this model, as well as the justification for this structure. To view the code itself, the github repository for this project can be found in the appendix.

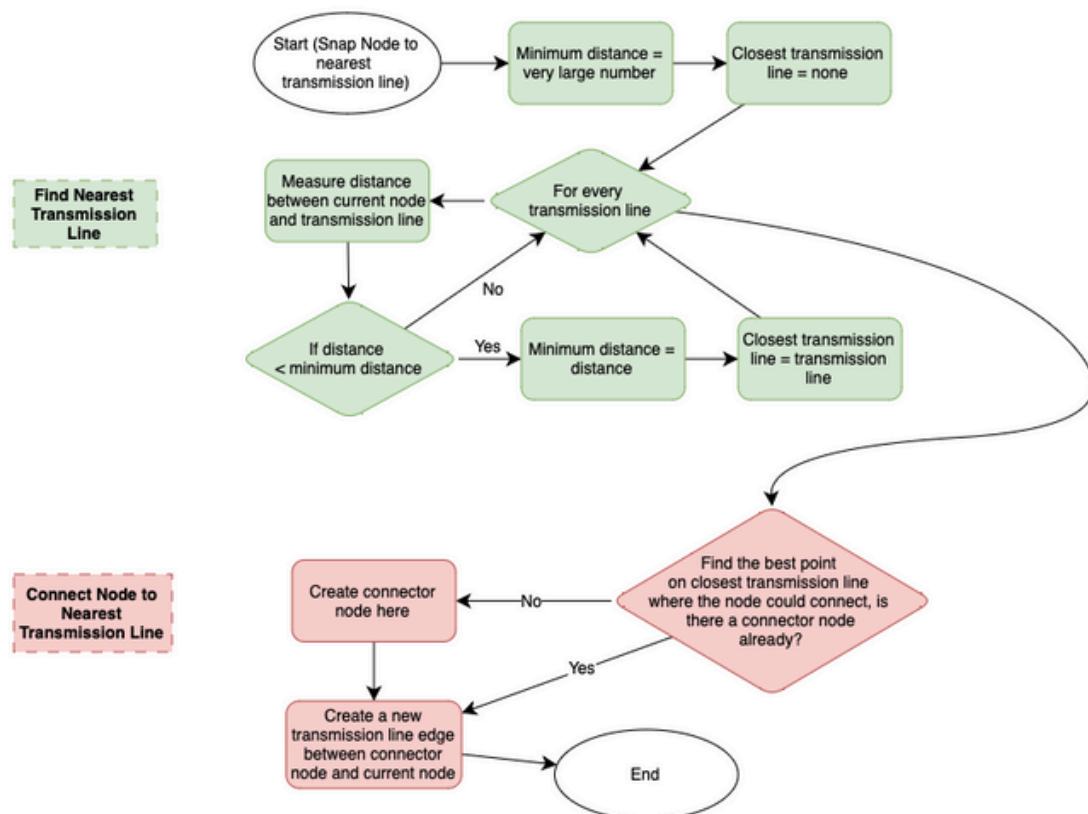
Note: Connector nodes / edges refer to nodes and edges that do not represent anything real in the data (targets, power plants, transmission lines), but are simply used to 'glue' the network together so that there aren't any gaps.



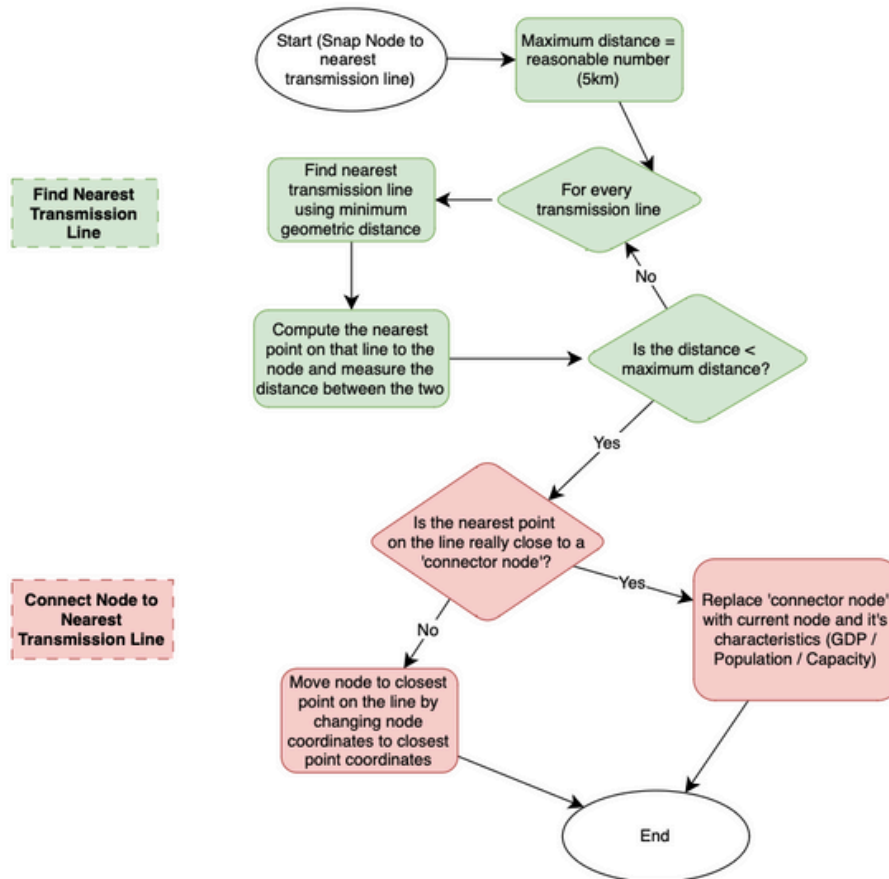
SAMPLE CODE OUTPUT AFTER 'HANDLING TARGETS' SECTION OF MAIN MODEL:



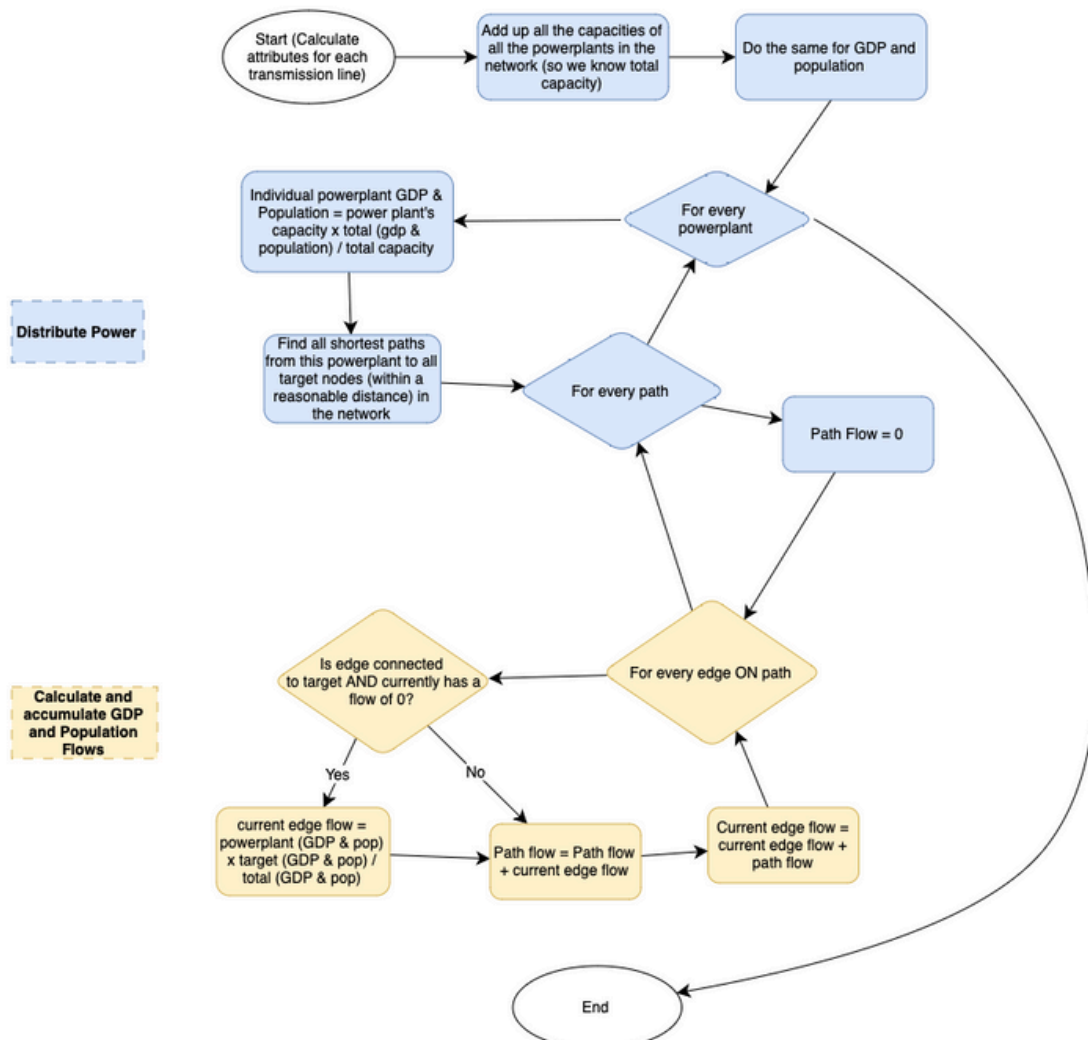
FIRST VERSION OF SNAPPING SUB-PROCESS:



SECOND VERSION OF SNAPPING SUB-PROCESS:



TRANSMISSION LINE ATTRIBUTION PROCESS:



Challenges

Challenge	Response
Memory and Time Consumption – India is very large, so running all of the data every time code is written or updated is incredibly time inefficient.	Running the code on selected regions of India whilst building and testing the model, and only running for the whole of India once we have full confidence in the functionality of the model.
Understanding the target binary raster file. The other data used for this project was fairly easy to understand, but the target data was unclear. It took extensive individual research and discussion with our client what a ‘target’ really meant. Additionally, as mentioned in the data section, the target file was formatted as a binary raster file, which was different from the regional target data that our reference code had.	We looked at the source data files to understand what a ‘target binary raster’ was, and how to use it in our model. However, because the source target data was formatted in this way, we had to leave it undisturbed, resulting in a very large quantity of target nodes. This exacerbated the aforementioned problem of data & computational overload when running large areas on our computers. Again, we addressed by running code on smaller areas and sending our client the workflow for larger areas.
Accounting for supply and demand logical error in the model’s methodology. The method we were given to assign power flow values to power lines assumes that every target region simply receives power from the powerplant that is nearest to it, which isn’t always true, especially in the case of a powerplant that has a small capacity and happens to be close to high power demand regions.	Two alternative methods integrated into our source material. 1) Calculate the amount of power a target needs and then logically decipher from which powerplant it must be receiving it from, rather than simply assuming every powerplant powers every target that it’s closest to. 2) Targets take power from the nearest powerplant. When the plant is empty, they move to the next closest powerplant.
Connecting target areas and powerplants to the electricity grid. The targets and powerplants coordinate data rarely matched perfectly onto the grid data, so various methods had to be used in order to infer the locations of where these connections are.	You can either ‘force’ the connection by moving existing nodes, or you can create ‘connector’ nodes and edges that branch off the grid to connect, without changing any of the original grid. We tried both methods and found that both work, the former is better in terms of computational load as you don’t need to make ‘connector’ nodes and edges, and the latter is slightly better in terms of preserving exact coordinate locations as you don’t move anything. Output variance between methods is minimal.

Challenge	Response
Using GDP vs GDP per capita Using GDP or GDP per capita would yield different results for our model. Both sources of data were associated with regions (regional GDP, vs regional GDP per capita). GDP regions were smaller and more numerous, and therefore more precise, than GDP per capita data. However, attributing GDP data to targets is a longer process.	We discussed with TF to see which was the preferred method. Using GDP per capita and multiplying by target population was concluded to be computationally simpler than the alternative. (Using GDP data, counting total population in a given region, dividing GDP by population, then multiplying said GDP per capita by each targets population.)
Assigning population and GDP values to targets. Our client informed us that ‘target’ data would need to have the following information: a) coordinate geometry (where it is) b) population of target and c) gdp. We were given 2 separate files to use to assign gdp to targets, so we had to figure out which one was the more appropriate to use.	The method we ended up using was using regional gdp per capita data, and multiplying by target populations to find individual target gdp. As mentioned earlier, this was to reduce computational complexity/load.
Methodology lacks transmission line electricity allocation. The base methodology only allocates electricity demand to targets; electricity flowing through transmission lines is not calculated.	We created new code to calculate electricity pathing from powerplants to targets - two methods were used. Method 1) Assume each target only consumes from one powerplant, calculate the combination of target-power plant links needed to provide power to all targets. Method 2) Each target consumes from the closest power plant. If the closest power plant is depleted, it consumes from the next closest plant.

Model Limitations

Simplified Flows of Electricity

Due to the laws of electricity and physics, transmission lines do not have unlimited carrying capacity and lose electricity over time due to resistance (Britannica). The model does not capture either phenomenon. Transmission lines are treated as having unlimited capacity, and losses during transmission are ignored. As a result, power flows in the model do not strictly obey the laws of electricity and should be understood as simplified estimates rather than physically realistic simulations. The lack of a capacity constraint per transmission line is the more fundamental issue, as it affects how electricity is routed and rerouted through the network following a climate hazard. Resistance losses are less important by comparison, as they primarily inflate quantities rather than change flow structure.

Relevance: This makes the model unsuitable for operational grid planning or load-balancing analysis. However, for climate risk modelling, where relative exposure and connectivity matter more than exact physical behaviour, this simplification is acceptable. Our contact at TF assured us that the model did not require such realism, as it would add unnecessary complexity and computational load.

Predicted transmission network geometry

Transmission lines are inferred using satellite-based nighttime light data and established open-source methodologies (Arderne, Zorn, Nicolas, et al., 2020). As the network is predicted rather than fully observed, spatial inaccuracies and missing lines may occur. These misalignments are a known trade-off in global and national-scale infrastructure modelling.

Relevance: While unsuitable for granular investment decisions (ie. a particular stretch of transmission line), the inferred network is sufficiently accurate for national-scale analysis where relative exposure and connectivity patterns are the primary concern. TF is aware that the model works in larger scales, and can advise on areas as a whole which are more or less vulnerable.

Electricity Supply Simplifications

1. In reality, electricity demand is dynamic, and met by a combination of sources distributed across the grid, the supply of which is also dynamic (Enbridge, 2024). However, the base methodology did not account for this dynamism, so our model is static, assuming constant values.
2. Our transmission line electricity allocation assumes uncapped transmission loads, nor does it incorporate the concept of resistance (electricity being lost as heat when travelling).
3. Method 1 of allocation assumes one plant per target, which is unrealistic. Method 2 assumes proximity is the overriding factor determining source, which is also unrealistic. A mix of electricity sources is often used, in proportion to the production of said source.

Relevance: Unlimited line capacity, paired with simplified sourcing of electricity, leads to certain transmission corridors appearing disproportionately important. This creates a conservative bias in vulnerability assessments, where individual connections appear more critical than they would be in practice. For risk screening and prioritisation, this bias is acceptable, but it limits the model's ability to capture redundancy and load-sharing.

Spatial aggregation of demand into target nodes

Target nodes represent large geographic areas rather than individual substations or consumers. For example, New Delhi is represented by 70 target nodes despite serving millions of people. Population and GDP are therefore condensed into single points, which reduces spatial realism.

Relevance: This aggregation is necessary for computational feasibility. Without it, the model would be intractable at the scale of India. However, it leads to an apparent redundancy of transmission lines, which in reality would distribute load across an area rather than serve a single point.

GDP is Population-Based

Economic activity is allocated to target nodes based on where people live, rather than where production occurs. Industrial zones, factories, and logistics hubs often generate significant GDP without corresponding resident populations, and these are underrepresented in the spatial distribution of demand.

Relevance: This limitation affects the realism of power-flow paths but is less consequential for aggregate exposure estimates. GDP per capita already distributes national economic output across the population, making this a defensible proxy at scale. However, this limits the models' ability to simulate industrial resilience. This is of particular note to investors who care about specific classes of industrial assets

Static Representation of GDP

GDP values are treated as fixed, average quantities. In reality, economic activity varies daily and seasonally, and may change significantly over years.

Relevance: A fixed GDP baseline is sufficient for long-term average risk assessment and comparison across regions. The model is not intended to capture short-term economic shocks or seasonal demand variation.

Exclusion of non-grid electricity sources

The model only accounts for electricity supplied by registered power plants in the dataset used. Distributed generation such as rooftop solar, backup generators, and microgrids are not included. This may cause areas with alternative power access to appear more vulnerable than they are in practice.

Relevance: This limitation may overstate risk in resilient regions but aligns with the model's focus on grid-dependent economic exposure. Capturing distributed generation would require data beyond the scope of the current project.

Outdated Datasets

GDP raster data is only available up to 2015, while population data extends to 2020. To maintain consistency, both metrics are anchored to the 2015 baseline, meaning results reflect historical rather than current economic conditions.

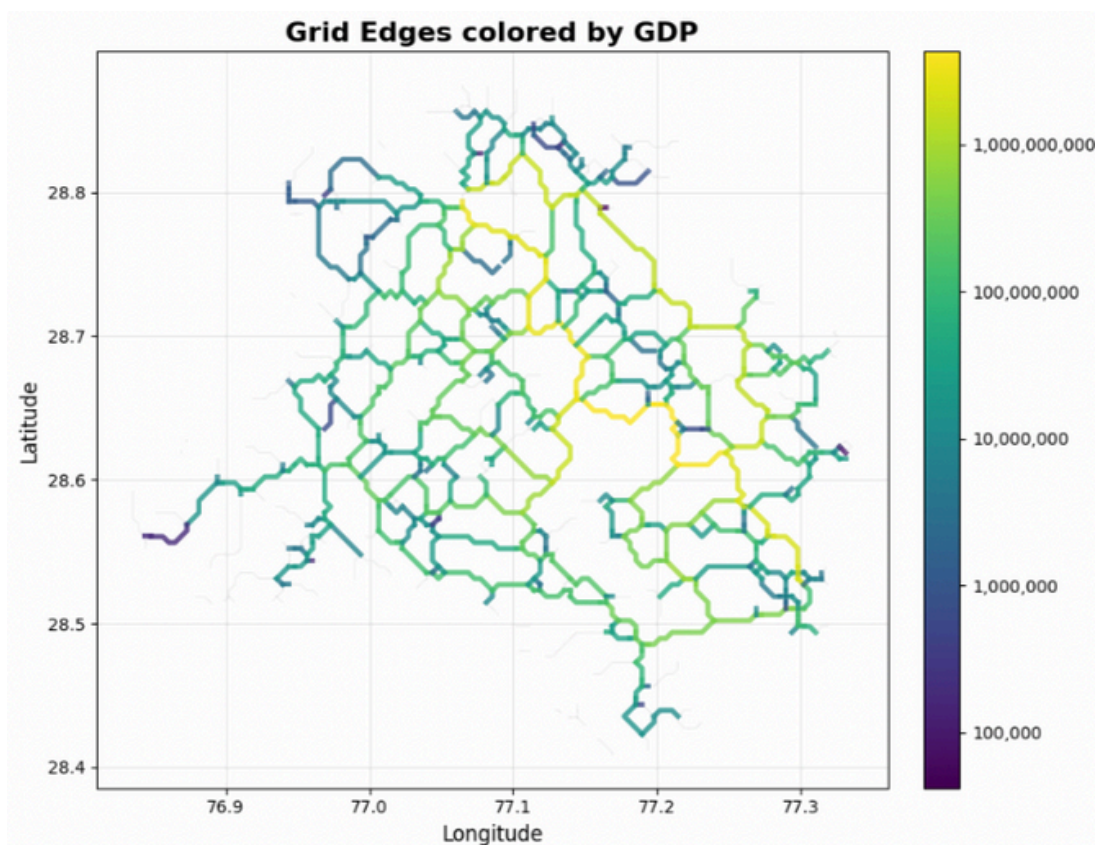
Relevance: This is a significant consideration for investors making present-day decisions. However, the limitation is data-driven rather than methodological; the model can be readily updated when newer datasets become available.

GDP-centric risk framing

The model prioritises GDP as the primary measure of exposure, reflecting a financial lens on climate risk. This approach may undervalue regions with low economic output but high human vulnerability, potentially reinforcing existing patterns of investment concentration.

Relevance: Appropriate for private-sector and financial clients focused on asset protection and returns. For public-sector or humanitarian applications, additional metrics capturing social vulnerability and human impacts would be required. TF professes to be a cascading risk company, therefore the risk of mass climate migration (cascading from a climate hazard) may be an additional factor they should account for in this and other models, as this could skew GDP distributions significantly.

SAMPLE OF THE FINAL OUTPUT (DELHI REGION)



Task 1 Conclusion

Overall, our work succeeded in producing a functioning electricity grid model for India. The resulting code was delivered to TF, who have access to greater computational resources and are therefore able to run the model at full national scale. While the assumptions and limitations of the model are worth acknowledging, they do not materially reduce its usefulness for TF's purposes. The primary objective was to capture connectivity and GDP exposure across the transmission network, rather than achieve fine-grained geographic or demographic precision. At this level, the model performs effectively and aligns with how TF and its clients intend to use it.

The most significant limitation concerns the representation of electricity flows. However, the allocation approach adopted mirrors that used in established professional projects such as Open GIRA, where precise electrical behaviour is secondary to modelling the structure of the network, the location of generation assets, and the distribution of demand. Other potential concerns, such as data granularity, were discussed with the client and deemed acceptable. As emphasised by TF, the climate-hazard risk model is intended as a tool for understanding and comparing risk, not as a fully realistic simulation of power system operations.

As a result, the project reduced development time for TF and demonstrated that their methodology can be successfully translated to new regional contexts. The work also helped surface the practical challenges involved in extending the model to a country as large and complex as India. With the electricity grid layer now rebuilt, TF is able to proceed with integrating it into the wider modelling pipeline and addressing outstanding issues related to visualisation and downstream analysis.

Scope and Goal

As many of TF's clients are invested in/ look to invest in infrastructure assets in developing assets, they are interested in seeing more granular asset-specific risk models that assess an asset's exposure to climate hazards.

In line with this, Group 2 was tasked with assessing the physical vulnerability of three hydropower plants in Uganda (Bugoye, Kikagati, and Nyamwamba II) to key climate hazards which could then be presented to TF's clients looking for an assessment or example of TF's more granular risk modelling. This involved developing asset-specific risk assessments that identify the relevant hazard types and their severity (e.g. droughts, flooding, and changes in precipitation and temperature), and evaluate their implications for physical performance, in particular electricity generation. While the broader modelling framework is designed to inform financial decision-making, Group 2's analysis focused on generation impacts, as asset-level financial data (e.g. revenues or cash flows) were not available and financial loss estimation was outside the scope of the task.

Hydropower Assets in Uganda

Hydropower assets are particularly sensitive to climate variability because their performance depends directly on hydrological conditions (Opperman et al, 2023). However, the mechanisms through which climate hazards affect generation differ substantially depending on plant type (run-of-river versus reservoir-based), location and hazard type. Hydropower plants generate electricity by converting river flows into energy, either instantaneously in run-of-river systems or through regulated storage in reservoir-based systems, implying fundamentally different exposure pathways to hydrological hazards (REPSOL, 2023). As a result, a one-size-fits-all risk metric is not appropriate.

The hazards considered in this analysis were:

- 1) Drought, affecting generation through reduced inflows;
- 2) Flooding, causing outages, downtime, or physical damage;
- 3) Chronic heat, potentially affecting efficiency
- 4) Sedimentation, affecting long-term storage capacity and turbine efficiency.

Conceptual Framework

Conceptual Framework: Hazards, Vulnerability, and Loss



A key conceptual distinction underlying this work is the separation between hazard, vulnerability, and loss. A hazard describes the physical intensity of a climate stressor; vulnerability captures how this hazard translates into physical damage and performance degradation, conditional on asset characteristics; in this case, increasing flood depth may raise the likelihood of damage to turbines, electrical equipment, or auxiliary infrastructure. Loss then refers to the resulting quantified reduction in electricity generation and associated revenue losses.

Assets Considered

- Kikagati: a reservoir-based dam, benefiting from storage buffering against short-term variability.
- Bugoye: a small run-of-river (RoR) plant with minimal storage, highly sensitive to hydrological variability.
- Nyamwamba II: a run-of-river plant located in a steep catchment, prone to rapid runoff and flash flooding.

These differences motivate the use of asset-specific vulnerability functions rather than uniform assumptions.

Location of the three selected hydropower plants in Uganda



Created with Datawrapper

Hazards Considered

Drought: Drought episodes are recurrent in Uganda and represent a major risk to hydropower generation, as electricity output depends directly on river inflows. Recent assessments indicate that under extreme dry conditions, Uganda's national hydropower generation could decline by up to ~90% over the next four decades (Mujjuni et al, 2023), highlighting the high sensitivity of Uganda's hydro-dependent power system to prolonged hydrological stress.

Riverine Flooding: Flooding poses a dual threat to hydropower.

- 1) Structural damage to physical components including dams, penstocks and turbines
- 2) Functional disruptions due to temporary debris blockage and increased sedimentation rates

Floods impact dam-type hydropower projects differently to run-off-river systems. For dams, the critical risks are overflowing and foundation defects (cracks in the dam) which account for roughly 85% of dam failures globally.(ASDS, 2020) Run-off-river systems lack buffers provided by reservoirs so are more likely to experience downtime due to floods with main risks being flash floods and increased debris flows. These differences require different approaches.

Chronic Heat: Chronic heat refers to periods of prolonged high ambient temperatures. Hydropower generators are typically designed for ambient temperatures $\leq 40^{\circ}\text{C}$ and water temperatures below 28°C at which point generation is impacted.(UNIDO, 2019) As temperature projections for Uganda do not project ambient temperatures to reach above $\leq 40^{\circ}\text{C}$, only increased water temperatures are considered as indicators of chronic heat risk.

Sedimentation: Sedimentation is the natural process where debris is carried by rivers and eventually settles in reservoirs or river beds creating build-up and thereby reducing water storage capacity, obstructing equipment and decreasing energy generation efficiency. Although sedimentation risk is not often considered in climate risk assessments, it poses a key challenge to hydropower production and the management of hydropower infrastructure with potential excess costs on investors. (Schellenberg et al, 2017) Due to limited data and academic literature on sedimentation's effect on run-off-river plant generational capacity only the dam-type plant Kikagati was considered.

Methodology

The methodology follows a common structure across hazards:

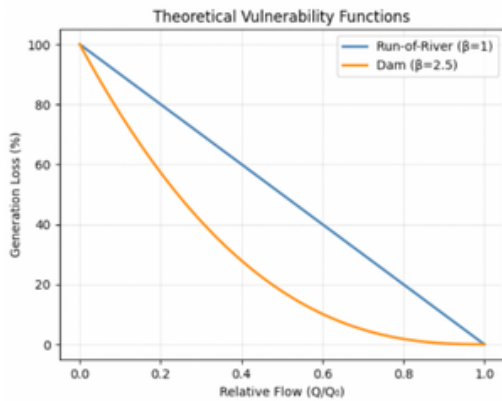
- 1) Hazard characterisation using open-source climate and hazard datasets (ERA5-Land, OS-Climate Physrisk, downscaled SSP projections). RCP 8.5 projections to 2050 were used as a high-emissions scenario to represent an upper-bound estimate of future climate conditions.
 - 2) Transformation of climate variables into asset-relevant hazard intensity metrics
 - 3) Application of vulnerability functions to map hazard intensity to generation loss.
 - 4) Aggregation of losses over time or event distributions to compute metrics such as Expected Annual Loss.
- Each hazard required a tailored approach, described in below and in detail in the Appendix.



Results and Implications

Drought Risk

- Vulnerability Function:



Run-of-river plants (Bugoye and Nyamwamba II) exhibit a linear vulnerability response, reflecting their lack of storage capacity. As inflow declines, generation losses increase proportionally. In contrast, the reservoir-based plant (Kikagati) exhibits a concave vulnerability response, capturing the buffering effect of storage, which allows moderate inflow deficits to be absorbed with limited generation impact.

- Precipitation-Based Drought Losses:

Projected precipitation-driven flow changes and generation losses

Site	Year	Baseline flow (Q_0)	Rainfall (mm)	Precipitation anomaly (%)	Flow ratio (Q/Q_0)	Generation loss (%)
Bugoye	2020	170.50	156.87	-7.99	0.83	16.75
Bugoye	2030	170.50	165.04	-3.20	0.93	6.91
Bugoye	2040	170.50	172.18	0.99	1.02	0.00
Bugoye	2050	170.50	187.51	9.98	1.23	0.00
Bugoye	2075	170.50	219.16	28.54	1.74	0.00
Kikagati	2020	38.47	62.38	62.15	1.97	0.00
Kikagati	2030	38.47	63.38	64.77	2.01	0.00
Kikagati	2040	38.47	65.16	69.38	2.09	0.00
Kikagati	2050	38.47	68.10	77.02	2.22	0.00
Kikagati	2075	38.47	76.94	99.99	2.64	0.00
Nyamwamba II	2020	112.92	153.94	36.33	1.98	0.00
Nyamwamba II	2030	112.92	160.73	42.35	2.17	0.00
Nyamwamba II	2040	112.92	166.87	47.78	2.36	0.00
Nyamwamba II	2050	112.92	181.17	60.45	2.83	0.00
Nyamwamba II	2075	112.92	211.59	87.39	3.98	0.00

Under precipitation-based projections, drought losses are limited and temporary. Bugoye experiences losses of $\approx 17\%$ (2020) and $\approx 7\%$ (2030) due to below-baseline precipitation, while from 2040 onward no losses occur. Kikagati and Nyamwamba II remain above baseline throughout, resulting in zero losses, reflecting the thresholded nature of the vulnerability function (losses occur only when $Q/Q_0 < 1$).

- SPEI-Based Drought Losses:

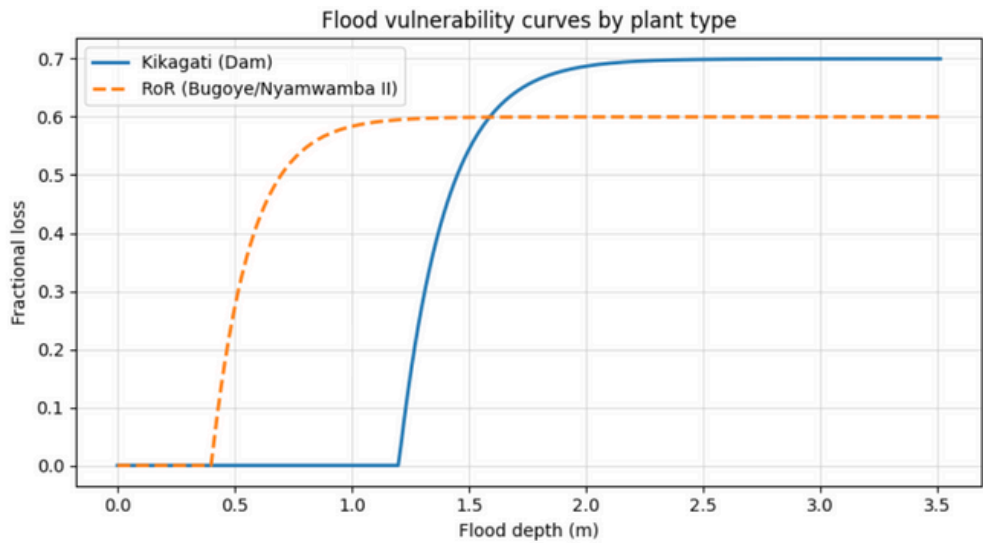
SPEI-based drought-related generation losses (%)

Site	2030	2040	2050	2080
Kikagati (Reservoir)	2.2%	2.4%	3.3%	2.4%
Bugoye (RoR)	4.0%	5.0%	7.7%	5.6%
Nyamwamba II (RoR)	4.0%	5.0%	7.7%	5.6%

Using SPEI-based drought frequency, generation losses remain non-zero even when annual precipitation is above baseline, capturing the importance of intra-annual drought timing. Run-of-river plants (Bugoye and Nyamwamba II) exhibit similarly high losses, reflecting both their limited buffering capacity and their close geographic proximity, which exposes them to comparable drought conditions. In contrast, the reservoir plant (Kikagati) is less affected.

Flood Risk

Plant	Threshold D_0 (m)	k	L_{max}	Depth scale (m)	EAL (%)	Qualitative interpretation
Bugoye (RoR)	0.4	6	0.6	0.9	≈ 6	Very exposed small RoR plant; shallow floods trigger downtime.
Kikagati (Dam)	1.2	5	0.7	1	≈ 4	Moderate. Reservoir buffer reduces damage frequency.
Nyamwamba II (RoR)	0.4	6	0.6	1	≈ 7	Steep Rwenzori catchment. High flood-related loss risk



EAL values reflect Bugoye and Nyamwamba II's higher vulnerability to flooding. Kikigati has a slightly lower EAL (4%) reflecting the buffering effect of a reservoir which has excess water storage capacity. Nyamwamba's EAL is slightly higher due to geomorphological conditions (steep terrain and rapid runoff).

The vulnerability functions shown above are threshold based meaning no damage occurs until enough water accumulates to effect the functioning of the plant. When this is reached losses accumulate exponentially until the maximum fractional loss of generation is reached.

Kikagati Monte Carlo Simulation

Due to the availability of a return period curve for Kikagati, an additional Monte Carlo simulation largest-event-per-year was carried out. For each-simulated year, the maximum flood depth was sampled from the exceedance distribution and converted into fractional generation loss. Annual losses are aggregated over 100,000 simulations.

Loss Distribution	
Metric	Value
Mean annual loss	3.83%
Maximum	70%
P(loss > 10%)	6.80%

Kikagati's flood risk profile is characterised by low-frequency, high-loss events. Mean annual loss is around 4%, however, the maximum annual loss could reach 70% although the chance of this maximum event occurring is only 6.8%. Over a 10-year horizon the Probability of a loss between 1 and 5% of annual generation is 47.8% over 20-years it is 72.8% . These results motivate adaptation measures to protect the client's investment from the long-term risk of flooding despite the relatively low mean annual loss value.

Chronic Heat

Results demonstrate a relatively low estimated annual generation loss across all three plants. Even if we look at multi-year loss projections over 100 years they never exceed 3.5% of annual generation for any asset. This suggests that none of the plants exhibit levels of vulnerability to chronic heat stress that require significant attention.

Multi-year Loss Due to Water Heat Stress

Plant	Estimated annual loss (%)	5yrs (%)	10yrs(%)	50yrs(%)	100 years(%)
Bugoye	0.0175	0.0875	0.175	0.875	1.75
Kikagati	0.0321	0.01605	0.321	0.1605	3.21
Nayawamba II	0.0135	0.0675	0.135	0.675	1.35

Results are shown across different time horizons in percentage loss of annual generation.

Linear vulnerability curves are constructed for a visual representation of results and included in the Appendix.

Sedimentation (Kikagati only)

Expected annual dam storage losses (0.63% (ICOLD, 2009)) were accumulated over time to estimate cumulative storage degradation. This yields projected capacity reductions of 3.15% over 5 years, 6.30% over 10 years, 12.59% over 20 years, and 31.48% over 50 years. This indicates significant long-term sedimentation risk to storage capacity for the Kikagati hydropower plant. Maintenance will be required and additional costs to cover this must be taken into account.

Time Horizon (years)	Cumulative Storage Loss
1	0.63%
5	3.15%
10	6.30%
20	12.59%
50	31.48%

However, we know that sedimentation does not occur linearly but is driven by flood events which mobilise sediment and carry it to reservoirs. To distribute the annual predicted capacity loss across different flood magnitudes a function was created to assess sediment mobilisation per flood event yielding the following results:

Flood Event	Per-Event Storage Loss
1-in-5 years	0.76%
1-in-10 years	1.44%
1-in-25 years	2.57%
1-in-50 years	3.58%

These values represent the per-event capacity loss for a 1-in-5, 10, 25 and 50 year floods. The values are small yet significant, specifically for floods of higher magnitudes. Overall, our results suggest a significant sedimentation risk that will likely need significant maintenance and/ or adaptation measures to ensure electricity production and limit sedimentation impact from high intensity floods which may become more frequent due to various climate change factors.

Challenges

This analysis faced several challenges, primarily driven by data constraints and the inherent complexity of modelling asset-level climate impacts on hydropower generation. Where possible, pragmatic modelling choices were made to ensure analytical consistency and relevance for client use.

Challenge	Response
Limited availability of asset-specific operational data: Plant-level operational data, historical generation profiles, flow–generation relationships, outage records, maintenance schedules, and adaptive operating practices were not available for the three hydropower assets analysed.	The analysis therefore focused on generation impacts inferred from physical exposure, using hazard intensity metrics combined with customised vulnerability functions rather than calibration to observed plant performance. Hence, results capture relative vulnerability across assets and hazards, but may not fully reflect plant-specific operational flexibility or short-term adaptive responses.
Absence of region- and asset-specific vulnerability curves: No empirically derived vulnerability curves exist for hydropower assets in Uganda, or for comparable assets in similar hydrological and climatic contexts. In particular, there is limited empirical evidence linking drought severity, flood depth, or heat stress directly to generation loss at the asset level.	Vulnerability functions were constructed using: asset typology (run-of-river vs. reservoir-based), engineering intuition regarding buffering and storage effects (research-backed), functional forms commonly applied in infrastructure risk modelling (e.g. WRI Aqueduct, OS-Climate Physrisk, IMIA guidance). Hence, these functions provide a conceptually sound but simplified representation of vulnerability. Results should therefore be interpreted as indicative of sensitivity, not precise forecasts of realised losses.
Reliance on secondary literature and parameter assumptions: Key parameters (rainfall–runoff elasticities, drought-month flow reductions, changes to efficiency at different water temperatures, flood damage thresholds, and, sediment mobilisation rates) could not be directly observed and had to be sourced from/assumed using sector literature.	Model outputs are sensitive to parameter choices, particularly for non-linear hazards. Sensitivity analysis would be a natural extension with additional time and or data.
Simplification of real-world operational dynamics: The model does not explicitly represent dynamic reservoir dispatch, short-term operational adaptations, emergency response measures, or recovery timelines following extreme events.	These dynamics were deliberately abstracted to prioritise transparent hazard-to-impact mapping and consistency across assets and hazard types. Hence, the framework is best suited for screening and comparative risk assessment, rather than detailed operational optimisation or real-time system simulation.

Limitations and Relevance

Limitation	Description	Relevance
Scenario-based exposure modelling rather than performance forecasting	Generation impacts are inferred from hazard intensity and asset-specific vulnerability functions to assess structural exposure, not calibrated to historical operational outcomes or used to forecast exact future generation trajectories.	This design choice aligns with TF's use case, enabling forward-looking, comparable risk screening across assets and geographies in data-constrained contexts. For TF and its clients, the value lies in identifying relative exposure, key risk drivers, and sensitivity to modelling assumptions. The framework is therefore well suited for further due-diligence prioritisation and scenario analysis, indicating where deeper, asset-specific analysis would be required.
Some models incomplete due to missing data (Flood and Sedimentation Risk)	In the case of flood risk the model could not be extended to RoR models due to missing return curve data. For sedimentation the lack of data and secondary literature means only Kikagati could be covered.	Although this presents a genuine limitation to this risk assessment of this specific case it does not limit the relevance and transferability of the methodology for TF's future asset-specific assessments. As additional hydrological and sediment data become available, the framework can be readily extended to incorporate RoR plants and additional climate hazards, enabling more comprehensive risk modelling.
Use of simplified, transferable vulnerability functions	Vulnerability functions are constructed using asset typology (ROR vs reservoir), engineering logic, and established literature, rather than site-calibrated empirical curves. This prioritises consistency and transferability across assets.	For TF, this is a feature rather than a weakness: it enables rapid deployment across new geographies and asset classes, which is critical for a consulting business operating under tight timelines and heterogeneous data availability. Site-specific refinement can be layered on later where clients require deeper analysis.
Comparative rather than exhaustive hazard coverage	Hazards are analysed consistently, using the same methodology, but individually, hence not fully integrating compound or cascading effects across hazards and the different power plants.	This design choice supports clarity and interpretability, essential for client communication and next steps. It allows TF to explain why an asset is vulnerable and through which pathway, before escalating to more complex dependency modelling.
Case-study nature of the task	The Uganda hydropower risk assessment is framed as a case study designed to demonstrate methodology rather than deliver a production-ready risk product.	From TF's perspective, the primary value lies in capability building: - Validating that the framework can be extended to asset-level hydropower - Demonstrating how different hazards and hazard characterisations affect outcomes - Generating a reusable template for future client work in energy and infrastructure

Task 2 Conclusion

This analysis was designed as a forward-looking, asset-level climate risk case study, aligned with Tipping Frontier's aim of developing more granular and decision-relevant risk models for infrastructure investors. Rather than producing point forecasts, the focus was on stress-testing how different hydropower assets respond to multiple climate hazards under consistent modelling assumptions. The analysis relied on open geospatial hazard data and transparent, asset-specific vulnerability functions.

Given the data and modelling constraints, the results should be interpreted comparatively across assets and hazard types, and as scenario-based rather than predictive. They are most useful for identifying key risk drivers and relative exposure, rather than estimating precise future generation outcomes. Within this framing, several robust patterns emerge. Run-of-river plants consistently appear more exposed to hydrological variability, particularly drought timing and flooding, due to their limited buffering capacity. Reservoir-based assets show greater resilience to short-term variability, but remain vulnerable to longer-term processes such as sedimentation. Across hazards, modelling choices—especially how drought is defined—have a material effect on estimated impacts, highlighting the importance of hazard characterisation in asset-level risk assessments.

From a client perspective, the value of this work lies less in the specific numerical outputs than in the capabilities it demonstrates. The case study shows how Tipping Frontier's framework can be extended to new geographies and asset types, how asset-level vulnerability can be constructed in data-constrained settings, and how alternative hazard representations can be compared systematically. This provides a practical template for screening operational risk, prioritising assets for further analysis, and structuring discussions with investors or asset owners around climate-related performance risks.

More broadly, the analysis illustrates that climate risk is not uniform across superficially similar assets, and that asset type, location, and hazard pathways matter. By making these differences explicit, the work supports Tipping Frontier's broader objective of enabling more informed risk management decisions, whether through targeted adaptation measures, portfolio rebalancing, or further financial analysis built on top of the physical vulnerability outputs developed here.

This project can be understood as back-end research and model development for Tipping Frontier. By adopting TF methodology and applying and adapting it to new geographic areas, both teams have enabled TF to engage clients in new markets and across different analytical scales. Team 1's model component provides substance that TF can show to clients interested in investing in India. Moreover, the component output is ready to be incorporated into a wider climate-hazard-risk model of India, as the electric grid layer; thereby Team 1 took TF a step further in achieving their long term goal of a climate risk 'universe'. Team 2's research could be directly used to advise clients with a stake in the functioning of Uganda's hydroelectric plants. More importantly, Team 2 pioneered the use of TF methodology for asset-specific analysis for hydroelectric power. TF now understands the benefits and drawbacks of their methodology, and obtained a proof-of-concept for their plans for asset-specific analysis. Cumulatively, our teams demonstrated that the TF framework is viable at large scale (Team 1: transplanting methodology to new countries) and small scale (Team 2: Asset-specific analysis).

In operational terms, the project increased TF's readiness to engage clients interested in Uganda and India. Prior to our work, much of the methodology existed as internal documentation, partial code, or conceptual frameworks that were not translated into demonstrable outputs. By producing working, explainable models in two active geographies, the project reduced uncertainty around what could be delivered within realistic time, data constraints, and computational limits. This, in turn, lowers the lead time required to move from client enquiry to initial analysis, and provides TF with concrete prototypes that can be used to scope engagements, set expectations, and communicate capabilities to prospective clients.

At the same time, the project clarified the appropriate interpretation of TF's modelling outputs. Both workstreams reinforced that the value of the framework lies less in precise prediction than in structured comparison and identifying vulnerabilities. The outputs are best understood as tools for indicating where hazards are likely to cause financial damage, which components of infrastructure warrant deeper analysis, and how sensitivity to hazards varies across locations or asset types. Used in this way, the models support informed decision-making without overstating their realism or precision, and align with TF's positioning of climate risk modelling as an input to judgement rather than a substitute for it.

Looking ahead, the work places TF in a stronger position to build iteratively on its existing framework. With a functioning electricity grid layer for India and a tested approach to asset-level analysis for hydropower, TF can now focus on integration, refinement, and extension rather than foundational development. This includes stitching individual components into a coherent end-to-end pipeline, improving visualisation and communication of results, and layering additional hazards or financial metrics where appropriate.

More broadly, the project illustrates how TF's climate-risk 'universe' can be expanded incrementally by linking country-level system coverage with targeted asset-level analysis, allowing macro and micro perspectives to be combined in response to different client needs. In this sense, the work not only delivered immediate outputs, but also helped pioneer a practical roadmap for how TF can continue to scale its modelling capabilities in a credible way.

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TASK 1:

GitHub Repository: <https://github.com/starrycarina/india-project>

TASK 2:

DROUGHT:

For our assets, exposure to drought was assessed using two complementary characterisations of hydrological stress in order to capture both inter-annual water availability and intra-annual drought timing.

First, annual precipitation projections (SSP5-8.5) were translated into relative changes in mean annual river inflow, capturing shifts in total water availability.

Second, SPEI-12 drought-frequency indicators were used to estimate the expected fraction of drought months per year, allowing drought impacts to arise from the timing and persistence of dry periods, even when annual precipitation remains above baseline levels.

A1. PRECIPITATION-BASED INFLOW GENERATION

Where:

P/P_0 : relative annual precipitation (SSP5-8.5 vs. 1981–2010 baseline)

Q/Q_0 : relative mean annual inflow

γ : asset-specific rainfall–runoff elasticity

Elasticity parameters (literature-based):

- Run-of-river: $\gamma=2.2$ (Bugoye, Nyamwamba II), (Schaeffli et al. (2007))
- Reservoir: $=1.4$ (Kikagati), (Berghuijs et al. (2014))

$$\frac{Q}{Q_0} = \left(\frac{P}{P_0} \right)^\gamma$$

A2. SPEI BASED INFLOW ESTIMATION

To capture intra-annual drought timing, an effective annual inflow ratio was constructed as:

$$F(p, t) = 1 - p_{dry}(1 - r)$$

Where:

p_{dry} : fraction of months with $SPEI < -1.5$

r : mean flow ratio during drought months

- Run-of-river: $r = 0.4$
- Reservoir: $r = 0.7$
- Source: OS-Climate Physrisk (SPEI-12)

A3. DROUGHT VULNERABILITY FUNCTIONS

Run-of-River plants:

$$L_D = \begin{cases} 1 - Q/Q_0, & Q/Q_0 < 1 \\ 0, & Q/Q_0 \geq 1 \end{cases}$$

Reservoir plants:

$$L_D = \begin{cases} (1 - Q/Q_0)^\beta, & Q/Q_0 < 1 \\ 0, & Q/Q_0 \geq 1 \end{cases} \quad \text{with } \beta = 2.5$$

RIVERINE FLOODING:

B1. FLOOD DEPTH-LOSS FUNCTION

d = flood depth
 $D0$ = damage threshold
 k = steepness

$$P_{\text{damage}}(D) = \frac{1}{1 + e^{-k(D-D_0)}}, \quad L(D) = L_{\text{max}}P_{\text{damage}}(D)$$

$P_{\text{damage}}(D)$ = Fractional loss (of annual generation) at given depth
 L_{max} = maximum fractional loss

For Bugoye and Nyamwamba II: For Bugoye and Nyamwamba II:
 $D0=0.4\text{m}$ $D0=1.2\text{m}$
 $k=6$ $k=5$
 $L_{\text{max}}=0.60$ $L_{\text{max}}=0.70$

B.2 MONTE CARLO SIMULATION PER-RETURN-PERIOD CONTRIBUTION TO EXPECTED ANNUAL LOSS (KIKIGATI)

RP (yrs)	Depth (m)	Exceed. Prob.	Marginal Prob.	Loss at Depth	Midpoint Loss	Contribution (%/vr)
2	0.131	0.5	0.3	0	0	0
5	0.662	0.2	0.1	0	0	0
10	1.002	0.1	0.06	0	0.238	1.43
25	1.428	0.04	0.02	0.476	0.564	1.13
50	1.737	0.02	0.01	0.652	0.671	0.67
100	2.04	0.01	0.006	0.69	0.694	0.42
250	2.439	0.004	0.002	0.699	0.699	0.14
500	2.741	0.002	0.001	0.7	0.7	0.07
1000	3.042	0.001	0.001	0.7	0.7	0.07

B.2 MONTE CARLO SIMULATION LARGEST EVENT PER YEAR SUMMARY (KIKIGATI)

Metric	Value
Mean annual loss	3.83%
Median	0%
90th percentile	0%
Maximum	69.99%
P(loss > 5%)	6.80%
P(loss > 10%)	6.80%

CHRONIC HEAT:

Base assumption: -0.15% efficiency per +1°C beyond 28°C of the water temperature. (UNIDO,2019) Plant efficiency loss is assumed to increase linearly with water temperature above 28°C.

$\alpha = 0.15\% \text{ per } 1^{\circ}\text{C}$ $\Delta T = \text{annual mean exceedance temperature above } 28^{\circ}\text{C } (^{\circ}\text{C})$

$L_{heat} = \alpha \times \Delta T$ $L_{heat} = \text{annual capacity loss}$

DATA SOURCE:

- PhysRisk water temperature exceedance curves
- Scenarios: 2030 and 2050 (HadGEM)
- Threshold bins: 27.5–30 °C and >30 °C

For each plant, exceedance days were extracted from PhysRisk intensity–threshold curves.

C1. DEGREE-DAY CONVERSION

As data in return curves is given per week per every 2.5°C threshold, this had to be converted into mean days per year in the 27.5°C-30°C bin and 30°C-32.5°C as temperatures do not exceed this.

- 1.25 °C represents mean exceedance in 27.5–30 °C bin
 - 3.5 °C represents mean exceedance above 30 °C
- $DD_{28} = (D_{27.5-30} \times 1.25) + (D_{>30} \times 3.5)$

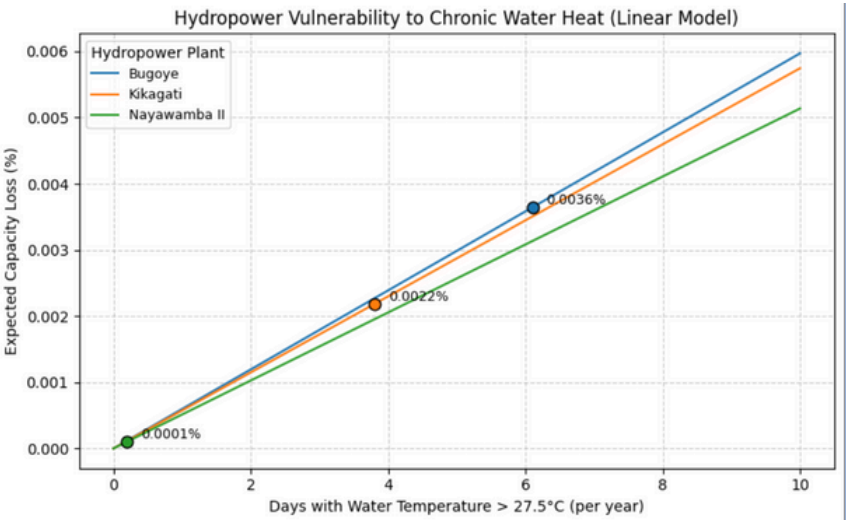
Annual mean temperature exceedance:

$\Delta T = DD_{28} / 365$

C2. APPLICATION OF ANNUAL CAPACITY LOSS FUNCTION (Per year and Cumulative)

Plant	$\Delta T (^{\circ}\text{C})$	Annual Heat Loss (%)	5 yrs	10 yrs	50 yrs	100 yrs
Bugoye	0.1165	0.0175	0.0875	0.175	0.875	1.75
Kikagati	0.2141	0.0321	0.1605	0.321	1.605	3.21
Nyamwamba II	0.0899	0.0135	0.0675	0.135	0.675	

C3. CHRONIC WATER HEAT LINEAR VULNERABILITY CURVES



SEDIMENTATION

D.1 SITE SPECIFIC GEOGRAPHIC CONDITIONS

Plant	Catchment Setting	Implication
Bugoye (Mubuku R.)	Mixed bedrock–alluvial downstream; alluvial floodplains	Moderate sediment deposition during floods
Kikagati (Kagera R.)	Steep relief, erodible soils, depleted vegetation	High sediment sensitivity
Nyamwamba II	Partially alluvial, dense river network, flash-flood prone	High geomorphic variability

D.2 LITERATURE JUSTIFICATION

Annual reservoir storage loss across the world ranges from 0.5–1.0% (White, 2001).

Literature for Uganda estimates this number to be closer to $\approx 0.63\%$ per year.(ICOLD,2009)

Sedimentation is strongly controlled by flood magnitude, rainfall variability, and channel morphology. .

D.3 SEDIMENT SCALING MODEL:

$$S_T = k \cdot Q_T^b$$

- S_T = sediment mass per flood event (t)
- Q_T = flood discharge for return period T (m³/s)
- k = proportionality constant, calibrated from a reference flood
- b = scaling exponent (controls sensitivity to flood size)

*The b scaling exponent demonstrates sensitivity to flooding. A lower b (below 1.5) means that sediment increases linearly with flood size and the sediment transport is dominated by frequent moderate floods. A higher b value means that sediment is dominated by rarer extreme floods, higher intensity causing more damage per-event. If the area has many flashy storms and landslide-prone areas a higher b is realistic.

In line with this a b value of 1.9 is chosen for Kikagati due to relatively high sediment sensitivity and literature including East African rating curves which identify b values between 1.7-2.2.

Tables showing results for both cumulative and per-event storage loss for Kikagati are included in the main report.